

## MCKINLEY NATL PARK, AK, USA

WMO: 702645

Lat: 63.733N

Lon: 148.917W

Elev: 524

StdP: 95.18

Time Zone: -9.00 (NAK)

Period: 95-19

WBAN: 46403

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |     |       | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|-----|-------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%  |       |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS  | MCDB  | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l) | (m)   | (n)                      | (o)  | (p)   |
| 1             | -40.9      | -37.5 | -42.1                         | 0.1 | -38.2 | -37.7 | 0.1 | -34.2 | 8.0                   | -10.6 | 7.3 | -10.5 | 0.0                      | 220  | 0.727 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.9                    | 23.7            | 13.7 | 22.1 | 13.2 | 19.9 | 12.4 | 15.0                | 21.4 | 13.9 | 19.8 | 12.9 | 18.4 | 3.1                  | 10   |

| Dehumidification DP/MCDB and HR |     |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 12.3                            | 9.5 | 16.3 | 11.3 | 8.9 | 15.4 | 10.2 | 8.3 | 14.7 | 43.4          | 21.6 | 40.4 | 20.0 | 37.9 | 18.2 | 18.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 6.9           | 5.8  | 5.3 | DB                         | -42.5 | 26.8               | 3.0 | 2.2                                                | -44.6 | 28.4       | -46.3 | 29.6       | -48.0 | 30.9       | -50.1 | 32.4 | (4) |
| (5)               |      |     | WB                         | -37.4 | 16.2               | 4.3 | 1.4                                                | -40.5 | 17.2       | -43.0 | 18.1       | -45.5 | 18.9       | -48.6 | 19.9 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | -2.1   | -17.6 | -12.3 | -11.2 | -1.7  | 6.2   | 11.8  | 13.2  | 10.9  | 5.6   | -2.5  | -12.6 | -15.7 | (6)  |  |
|                                                                        | DBStd       | 13.19  | 12.36 | 9.54  | 8.04  | 5.78  | 3.89  | 2.82  | 2.59  | 3.15  | 4.08  | 6.43  | 9.26  | 10.63 | (7)  |  |
|                                                                        | HDD10.0     | 4659   | 856   | 624   | 658   | 351   | 127   | 13    | 4     | 26    | 138   | 390   | 677   | 796   | (8)  |  |
|                                                                        | HDD18.3     | 7463   | 1114  | 858   | 916   | 601   | 376   | 197   | 159   | 232   | 383   | 647   | 927   | 1054  | (9)  |  |
|                                                                        | CDD10.0     | 239    | 0     | 0     | 0     | 0     | 10    | 66    | 104   | 53    | 5     | 1     | 0     | 0     | (10) |  |
|                                                                        | CDD18.3     | 1      | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0     | 0     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 72     | 0     | 0     | 0     | 0     | 2     | 22    | 32    | 17    | 0     | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 6      | 0     | 0     | 0     | 0     | 0     | 2     | 3     | 1     | 0     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 1.8    | 1.5   | 1.7   | 2.2   | 2.3   | 2.2   | 1.9   | 1.7   | 1.7   | 1.8   | 1.9   | 1.5   | 1.5   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 424    | 22    | 19    | 13    | 14    | 20    | 50    | 73    | 79    | 56    | 27    | 25    | 24    | (15) |  |
|                                                                        | PrecMax     | 672    | 72    | 54    | 47    | 65    | 48    | 102   | 169   | 141   | 143   | 66    | 61    | 101   | (16) |  |
|                                                                        | PrecMin     | 328    | 4     | 4     | 1     | 2     | 6     | 19    | 25    | 34    | 18    | 9     | 5     | 4     | (17) |  |
|                                                                        | PrecStd     | 81     | 17    | 12    | 10    | 14    | 13    | 23    | 32    | 30    | 33    | 14    | 17    | 22    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 7.4   | 7.4   | 7.7   | 15.9  | 22.8  | 26.2  | 26.6  | 25.3  | 17.6  | 14.0  | 7.6   | 7.0   | (19) |  |
|                                                                        |             | MCWB   | 3.7   | 2.0   | 2.4   | 7.2   | 11.3  | 13.9  | 15.0  | 14.8  | 10.0  | 7.7   | 3.2   | 2.3   | (20) |  |
|                                                                        | 2%          | DB     | 4.1   | 4.0   | 4.9   | 11.3  | 19.8  | 22.8  | 23.8  | 22.5  | 16.0  | 11.0  | 5.0   | 3.9   | (21) |  |
|                                                                        |             | MCWB   | 1.2   | 0.7   | 1.0   | 4.5   | 10.4  | 12.8  | 14.1  | 14.0  | 9.4   | 6.5   | 1.8   | 0.8   | (22) |  |
|                                                                        | 5%          | DB     | 2.6   | 2.4   | 2.7   | 8.8   | 17.3  | 21.2  | 22.1  | 19.0  | 13.9  | 7.9   | 2.8   | 2.4   | (23) |  |
|                                                                        |             | MCWB   | 0.1   | -0.7  | -0.7  | 3.3   | 8.6   | 12.3  | 13.7  | 12.7  | 8.4   | 4.2   | 0.3   | -0.4  | (24) |  |
|                                                                        | 10%         | DB     | 0.2   | 0.8   | 1.2   | 7.3   | 14.7  | 18.9  | 19.9  | 17.2  | 12.5  | 6.6   | 1.1   | 0.0   | (25) |  |
|                                                                        |             | MCWB   | -1.9  | -1.9  | -1.9  | 2.2   | 7.4   | 11.1  | 13.1  | 11.5  | 7.9   | 3.2   | -0.8  | -2.0  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 4.1   | 2.7   | 3.1   | 7.2   | 11.9  | 15.8  | 16.6  | 16.1  | 11.4  | 8.8   | 3.6   | 2.8   | (27) |  |
|                                                                        |             | MCDB   | 7.2   | 5.8   | 6.7   | 15.5  | 19.8  | 22.6  | 24.6  | 23.5  | 16.1  | 13.5  | 7.1   | 6.1   | (28) |  |
|                                                                        | 2%          | WB     | 2.0   | 1.1   | 1.2   | 4.8   | 10.6  | 14.1  | 15.1  | 14.5  | 10.1  | 6.4   | 2.0   | 1.2   | (29) |  |
|                                                                        |             | MCDB   | 4.3   | 4.5   | 4.6   | 10.0  | 19.3  | 20.9  | 21.6  | 20.6  | 14.4  | 10.3  | 4.9   | 3.7   | (30) |  |
|                                                                        | 5%          | WB     | 0.4   | -0.5  | -0.4  | 3.6   | 9.1   | 12.8  | 14.1  | 13.1  | 9.2   | 4.5   | 0.6   | -0.3  | (31) |  |
|                                                                        |             | MCDB   | 2.9   | 2.1   | 2.6   | 8.2   | 16.2  | 19.4  | 20.0  | 18.2  | 13.3  | 7.8   | 2.8   | 2.4   | (32) |  |
|                                                                        | 10%         | WB     | -1.5  | -2.1  | -1.9  | 2.4   | 7.9   | 11.8  | 13.3  | 12.0  | 8.1   | 3.4   | -1.0  | -2.0  | (33) |  |
|                                                                        |             | MCDB   | 0.6   | 0.2   | 0.9   | 6.9   | 13.9  | 17.6  | 18.8  | 15.9  | 11.7  | 6.3   | 1.0   | 0.0   | (34) |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 9.5   | 11.9  | 14.1  | 11.7  | 11.4  | 11.2  | 9.9   | 9.2   | 9.2   | 8.6   | 8.9   | 9.3   | (35) |  |
|                                                                        | 5% DB       | MCDBR  | 8.9   | 10.8  | 12.3  | 14.1  | 17.2  | 16.0  | 15.9  | 14.5  | 12.9  | 9.3   | 8.2   | 9.1   | (36) |  |
|                                                                        |             | MCWBR  | 6.8   | 8.3   | 8.8   | 8.4   | 9.0   | 7.8   | 7.8   | 7.5   | 7.6   | 5.9   | 6.2   | 7.2   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 9.0   | 10.0  | 10.5  | 12.7  | 15.3  | 13.6  | 12.8  | 12.1  | 11.1  | 8.3   | 7.8   | 8.8   | (38) |  |
|                                                                        |             | MCWBR  | 7.1   | 8.0   | 7.4   | 7.7   | 8.1   | 7.0   | 6.7   | 6.6   | 6.6   | 5.5   | 6.0   | 7.2   | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.193 | 0.261 | 0.269 | 0.321 | 0.357 | 0.391 | 0.366 | 0.380 | 0.306 | 0.262 | 0.194 | 0.185 | (40) |  |
|                                                                        | taud        |        | 1.959 | 2.123 | 2.219 | 2.134 | 2.250 | 2.258 | 2.356 | 2.316 | 2.491 | 2.318 | 1.942 | 1.899 | (41) |  |
|                                                                        | Ebn at Noon |        | 410   | 659   | 832   | 850   | 850   | 824   | 839   | 785   | 796   | 671   | 382   | 176   | (42) |  |
|                                                                        | Edh at Noon |        | 29    | 62    | 86    | 118   | 117   | 119   | 105   | 100   | 68    | 50    | 27    | 13    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.19  | 0.87  | 2.68  | 4.62  | 5.34  | 5.24  | 4.51  | 3.50  | 2.26  | 0.99  | 0.29  | 0.08  | (44) |  |
|                                                                        | RadStd      |        | 0.02  | 0.09  | 0.22  | 0.28  | 0.47  | 0.37  | 0.35  | 0.32  | 0.31  | 0.13  | 0.04  | 0.01  | (45) |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (7 neighbors) |  | +0.83 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -314    | N/A     | N/A     |

Nomenclature: See separate page